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**Press Information Bureau
Government of India
Ministry of Jal Shakti**

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Flood Forecasting and Early Warning System

Forecasting system is essential to predict likely increase in water level in rivers. Central Water Commission (CWC) provides statistical methodology (gauge to gauge correlation) based short range flood forecast. Presently, flood forecasts are issued by CWC at 332 stations (133 Inflow Forecast Stations and 199 Level Forecast Stations). Annually, about 10,000 flood forecasts are issued by CWC.

CWC is currently providing near real time five-day advisory flood forecast on its web portal <https://aff.india-water.gov.in/> for 20 major river basins of the country. The five day advance forecast is generated using various available rainfall data products like forecast rainfall data GFS (Global Forecast System) and WRF (Weather Research and Forecasting) provided by IMD (Indian Meteorological Department), rainfall estimates namely GsMaP (Global Satellite Mapping of Precipitation) & GPM (Global Precipitation Measurement), as a major input into the model system. The system is developed in-house using modelling software for flood forecasting which is updated every three hours for all the stations simultaneously in automatic mode during monsoon season.

CWC has also taken up the work of development of near real time Inundation Forecast for Ganga Basin through consultancy under National Hydrology Project (NHP) using High Resolution Digital Elevation Model (DEM) received from NRSC/ Survey of India / State Govt.

The data collected from field stations is transmitted from the site to the concerned Flood Forecasting Centre of CWC through wireless and/or telephone/mobile and satellite based telemetry system & VSAT. Central Water Commission maintains Wireless Stations for near real-time data communication. These wireless sets work on pre-fixed schedules for receiving the vital hydro-meteorological data immediately after its observation. Now under modernization program, satellite based Telemetry System has been installed at various stations for sensor based automatic data collection and satellite based communication. The data is transmitted to the Earth Receiving Station through Satellite and then to various Modelling Centres through VSAT Systems. The nodal officer of the dam/reservoir share reservoir related data with CWC through uploading on Water Information Management System (WIMS) or sending through e-mail/SMS/Phone/Wireless etc.

Dissemination of flood forecasts has also been modernised by having a dedicated website <https://ffs.india-water.gov.in>. In addition, telephone/mobile phone, fax and internet are used for dissemination of flood forecasts to user agencies. Daily Flood Situation Reports cum Advisories are shared with all stakeholders as well as general public using social media platforms.

This information was given by the Minister of State for Jal Shakti, Shri Bishweswar Tudu in a written reply in Rajya Sabha today.

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